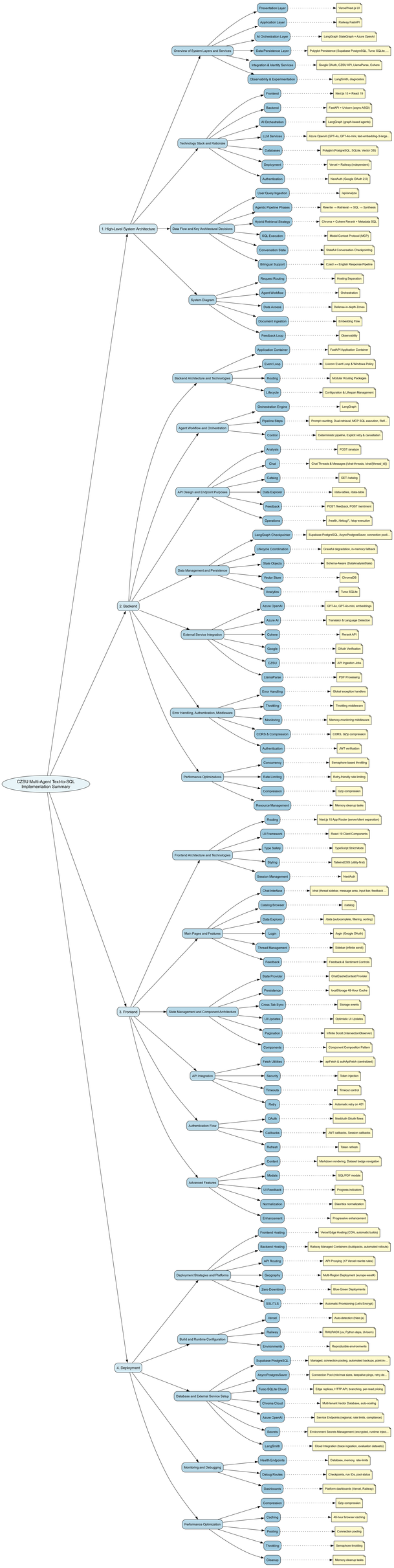
CZSU Multi-Agent Text-to-SQL Implementation Summary
1. High-Level System Architecture
Overview of System Layers and Services
Presentation Layer
Vercel Next.js UI
Application Layer
Railway FastAPI
AI Orchestration Layer
LangGraph StateGraph + Azure OpenAI
Data Persistence Layer
Polyglot Persistence (Supabase PostgreSQL, Turso SQLite,...
Integration & Identity Services
Google OAuth, CZSU API, LlamaParse, Cohere
Observability & Experimentation
LangSmith, diagnostics
Technology Stack and Rationale
Frontend
Next.js 15 + React 19
Backend
FastAPI + Uvicorn (async ASGI)
AI Orchestration
LangGraph (graph-based agents)
LLM Services
Azure OpenAI (GPT-4o, GPT-4o-mini, text-embedding-3-large;...
Databases
Polyglot (PostgreSQL, SQLite, Vector DB)
Deployment
Vercel + Railway (independent)
Authentication
NextAuth (Google OAuth 2.0)
Data Flow and Key Architectural Decisions
User Query Ingestion
/api/analyze
Agentic Pipeline Phases
Rewrite → Retrieval → SQL → Synthesis
Hybrid Retrieval Strategy
Chroma + Cohere Rerank + Metadata SQL
SQL Execution
Model Context Protocol (MCP)
Conversation State
Stateful Conversation Checkpointing
Bilingual Support
Czech ↔ English Response Pipeline
System Diagram
Request Routing
Hosting Separation
Agent Workflow
Orchestration
Data Access
Defense-in-depth Zones
Document Ingestion
Embedding Flow
Feedback Loop
Observability
2. Backend
Backend Architecture and Technologies
Application Container
FastAPI Application Container
Event Loop
Uvicorn Event Loop & Windows Policy
Routing
Modular Routing Packages
Lifecycle
Configuration & Lifespan Management
Agent Workflow and Orchestration
Orchestration Engine
LangGraph
Pipeline Steps
Prompt rewriting, Dual retrieval, MCP SQL execution, Refi...
Control
Deterministic pipeline, Explicit retry & cancellation
API Design and Endpoint Purposes
Analysis
POST /analyze
Chat
Chat Threads & Messages (/chat-threads, /chat/{thread_id})
Catalog
GET /catalog
Data Explorer
/data-tables, /data-table
Feedback
POST /feedback, POST /sentiment
Operations
/health, /debug/*, /stop-execution
Data Management and Persistence
LangGraph Checkpointer
Supabase PostgreSQL, AsyncPostgresSaver, connection pool...
Lifecycle Coordination
Graceful degradation, in-memory fallback
State Objects
Schema-Aware (DataAnalysisState)
Vector Store
ChromaDB
Analytics
Turso SQLite
External Service Integration
Azure OpenAI
GPT-4o, GPT-4o-mini, embeddings
Azure AI
Translator & Language Detection
Cohere
Rerank API
Google
OAuth Verification
CZSU
API Ingestion Jobs
LlamaParse
PDF Processing
Error Handling, Authentication, Middleware
Error Handling
Global exception handlers
Throttling
Throttling middleware
Monitoring
Memory-monitoring middleware
CORS & Compression
CORS, GZip compression
Authentication
JWT verification
Performance Optimizations
Concurrency
Semaphore-based throttling
Rate Limiting
Retry-friendly rate limiting
Compression
Gzip compression
Resource Management
Memory cleanup tasks
3. Frontend
Frontend Architecture and Technologies
Routing
Next.js 15 App Router (server/client separation)
UI Framework
React 19 Client Components
Type Safety
TypeScript Strict Mode
Styling
Tailwind CSS (utility-first)
Session Management
NextAuth
Main Pages and Features
Chat Interface
/chat (thread sidebar, message area, input bar, feedback ...
Catalog Browser
/catalog
Data Explorer
/data (autocomplete, filtering, sorting)
Login
/login (Google OAuth)
Thread Management
Sidebar (infinite scroll)
Feedback
Feedback & Sentiment Controls
State Management and Component Architecture
State Provider
ChatCacheContext Provider
Persistence
localStorage 48-Hour Cache
Cross-Tab Sync
Storage events
UI Updates
Optimistic UI Updates
Pagination
Infinite Scroll (IntersectionObserver)
Components
Component Composition Pattern
API Integration
Fetch Utilities
apiFetch & authApiFetch (centralized)
Security
Token injection
Timeouts
Timeout control
Retry
Automatic retry on 401
Authentication Flow
OAuth
NextAuth OAuth flows
Callbacks
JWT callbacks, Session callbacks
Refresh
Token refresh
Advanced Features
Content
Markdown rendering, Dataset badge navigation
Modals
SQL/PDF modals
UI Feedback
Progress indicators
Normalization
Diacritics normalization
Enhancement
Progressive enhancement
4. Deployment
Deployment Strategies and Platforms
Frontend Hosting
Vercel Edge Hosting (CDN, automatic builds)
Backend Hosting
Railway Managed Containers (buildpacks, automated rollouts)
API Routing
API Proxying (17 Vercel rewrite rules)
Geography
Multi-Region Deployment (europe-west4)
Zero-Downtime
Blue-Green Deployments
SSL/TLS
Automatic Provisioning (Let's Encrypt)
Build and Runtime Configuration
Vercel
Auto-detection (Next.js)
Railway
RAILPACK (uv, Python deps, Uvicorn)
Environments
Reproducible environments
Database and External Service Setup
Supabase PostgreSQL
Managed, connection pooling, automated backups, point-in-...
AsyncPostgresSaver
Connection Pool (min/max sizes, keepalive pings, retry de...
Turso SQLite Cloud
Edge replicas, HTTP API, branching, per-read pricing
Chroma Cloud
Multi-tenant Vector Database, auto-scaling
Azure OpenAI
Service Endpoints (regional, rate limits, compliance)
Secrets
Environment Secrets Management (encrypted, runtime inject...
LangSmith
Cloud Integration (trace ingestion, evaluation datasets)
Monitoring and Debugging
Health Endpoints
Database, memory, rate-limits
Debug Routes
Checkpoints, run IDs, pool status
Dashboards
Platform dashboards (Vercel, Railway)
Performance Optimization
Compression
Gzip compression
Caching
48-hour browser caching
Pooling
Connection pooling
Throttling
Semaphore throttling
Cleanup
Memory cleanup tasks